

Prairie AeroSurface Components LLC — Plant 3

South Wichita, Kansas · 2026-05-14 · Field Brief · Internal handover

● What this is

Customer frames it as 'three abnormal events'. The composite shows exceedances on seven of nine parameters — chronic envelope, not slug events.

Aerospace metal-finishing facility, indirect discharger to municipal POTW. Categorical user under 40 CFR Part 433. The WW-01 24-hour composite shows hex chrome at 3.4×, nickel at 3.1×, zinc at 3.0×, cyanide at 1.72× local limits. **The reframe is the deal.** If 'events' framing wins, this becomes a \$300K metals-precipitation point-fix; if the chronic-envelope reframe wins, it's a multi-million-dollar pretreatment redesign with PFAS source-control as phase-2.

● What we'd propose

Pretreatment redesign — Cr⁶⁺ reduction + cyanide destruction + EQ slug control + phase-2 PFAS source-control. \$3–8M over 18 months. Their \$300K alternative costs them ~\$4–7M over 5 years when surcharges, the forced PFAS retrofit, and risk-adjusted enforcement exposure are priced in.

Recommended approach: redesigned pretreatment train — Cr⁶⁺ reduction (bisulfite, upstream of hydroxide), two-stage alkaline-chlorination cyanide destruction with segregation from acid streams, hydroxide precipitation re-tuned for the Ni/Zn/Cu mix, and EQ-tank slug control (level interlocks + dump-discipline SOP + continuous pH trim). ~\$2–4M capex + design fee. **Phase-2:** PFAS source-control on the chromate conversion line + F006 residuals strategy. ~\$1–3M, 12–18 months out.

Why the customer should want this — the win-win argument:

Permanent compliance, not a 12-month band-aid. A redesigned train sized to the actual chemistry takes Prairie from chronic non-compliance to permanent compliance across all 7 parameters. Eliminates ongoing surcharges (~\$120K/yr at observed mass loading), takes them off the POTW's repeat-violator track, and ends the operator burden of running a train that was never sized for this chemistry.

PFAS handled before it becomes a forced retrofit. 6:2 FTS dominance + tightening federal effluent guidelines for chrome platers mean aerospace finishers will be required to source-control within 24–36 months. Doing it now as committed phase-2 is ~40% cheaper than a forced retrofit and avoids a second plant shutdown.

HCN risk neutralised. Segregating cyanide destruction from acid streams closes the active HCN-evolution pathway — material reduction in OSHA exposure and the underlying criminal-liability tail.

Cost of the alternative — fully priced over 5 years:

Cost component	Their \$300K point-fix path	Our proposal
Capex (year 0)	\$300K — metals precipitation only	\$2–4M — full train + EQ control
Sewer-use surcharges, 5yr	~\$600K — 4 of 7 params still exceed	~\$0 — back inside limits
Forced PFAS retrofit, yr 2–3	\$1.5–2.5M — at higher emergency cost	Included in phase-2 (\$1–3M)
Risk-adjusted enforcement	\$500K–1.5M — NOV + Consent Order tail	~\$50K — minimal residual
HCN incident exposure	Material, uncapped — open pathway	Neutralised by segregation
5-year total (mid-range)	~\$4–7M + HCN tail risk	~\$3–7M, risk extinguished

Deal-size sensitivity (within our \$3–8M range): scope of redesign drives ~3× from point-fix to full train; committed phase-2 PFAS adds ~\$2M + 12mo; recurring monitoring/SOP/training contract adds \$200–500K/yr. Confirming plant flows (action #2) tightens the range.

● What could kill it

The framing battle is the #1 kill risk. If 'three events' wins on Thursday, you lose this deal to a \$300K point-fix vendor.

1 The customer's 'events' framing wins

If the environmental manager pushes the slug-event story in the conversation and you don't reframe to chronic, a metals-precipitation vendor closes a point-fix at 1/10 the scope. *Mitigation: Open Thursday with "I've looked at the WW-01 composite — this isn't three events," and walk them through the 7-parameter exceedance directly.*

2 POTW enforcement posture goes hot or cold

Formal NOV or Consent Order spikes urgency but may force a fast point-fix; informal posture drops deal pace and the deal can drift 12+ months. *Mitigation: Get the city's posture in writing within 7 days; size the deal to whichever way it lands.*

3 HCN safety incident before close

Legacy cyanide line + documented EQ-tank pH 4.2 = active HCN-evolution pathway. An incident routes the conversation to emergency response and a different vendor. *Mitigation: Surface the safety angle Thursday explicitly — credibility builder, protects against incident-driven loss.*

● Do this next

Thursday's reframe call is the single highest-leverage move. Everything else calibrates around it.

1 Call the environmental manager with the chronic-envelope reframe · Thursday

Open with "I've looked at the WW-01 composite — this isn't three events, it's a chronic discharge envelope." Lead the conversation; don't accept their framing. Request the Playbook separately for the question structure.

2 Pull the ECHO record for the receiving POTW · by Friday

Public data, 30 minutes of work. Tells you the city's enforcement track record, prior NOV's, recent consent orders. Calibrates regulatory urgency.

3 Identify the corporate EHS approver for capex above \$500K · by next Monday

A \$3–8M deal needs corporate EHS sign-off. Find them before you're 4 conversations deep — common deal-killer is reaching approval and finding the approver was never engaged.